Data Manipulation Language

- Data Manipulation Language(DML) is a structured query language which is used for inserting, updating and deleting data in the database objects like table or view.
- DML comprises the SQL change statements which modifies the stored data but not the database object.
- DML consist of three SQL statements namely
  - Insert
  - Update
  - Delete

INSERT Statement

- Insert statement is used for inserting data into table.
- Insertion of data can be done in multiple ways.

Syntax:
INSERT INTO table_name[(column1, column2,...)]
VALUES(value1, value2,....);

```
SQL> insert into Phone_Book(Name,Phone_no,City)
  2  values('Ritika',9997123456,'Mdl');

1 row created.
```

If values in all the columns inserted in proper order, column names are not mandatory.

Syntax-:

INSERT INTO table_name VALUES(value1, value2,....);

```
SQL> insert into Phone_Book
  2  values('Subodh',9898989898,'hdr');

1 row created.
```

Insert Multiple Rows:

We can insert multiple rows using a **single** INSERT statement:

Syntax:

INSERT INTO table_name (col1,col2,..) VALUES

(.....values for record 1),

(.....values for record 1),

.....

(values for record n);

Example:

```
INSERT INTO cities (city_name, state) VALUES
    ('Guwahati', 'Assam'),
    ('Shillong', 'Meghalaya'),
    ('Imphal', 'Manipur');
```

Note: In the video available in this course , the syntax used to insert multiple rows to a table is supported by Oracle not by MySQL.

UPDATE statement

- Update command is used to change or modify data of one or more records in a table.

Syntax:

UPDATE Table_name SET Column_name1=value1
[,Column_name2=value2,...]
[WHERE Condition];

- All rows will be updated if no condition is specified in where clause.

```
SQL> select * from Phone_book;

NAME         PHONE_NO CITY
---------- ---------- ----------
Raj          349235745 Hyd
Nish         807773435 Del
Subodh     9897999999 Hdr
Brijesh    9595959595 Ngp

SQL> update Phone_Book
  2  set city='Mum'
  3  where name='Raj';

1 row updated.

SQL> select * from Phone_Book;

NAME         PHONE_NO CITY
---------- ---------- ----------
Raj          349235745 Mum
Nish         807773435 Del
Subodh     9897999999 Hdr
Brijesh    9595959595 Ngp
```

DELETE statement

- Delete statement is used to remove one or more records from a table.
- A subset may be defined for deletion using a condition, otherwise all records are removed.

Syntax:

DELETE FROM Table_Name [WHERE Condition];

Delete statement using WHERE condition.

```
SQL> delete from Phone_Book
  2  where phone_no=9595959595;

1 row deleted.

SQL> select * from Phone_Book;

NAME         PHONE_NO CITY
---------- ---------- ----------
Subodh     9897999999 hdr
```

Note: Delete statement without where condition deletes all the rows from table.

Example

```
SQL> select * from Phone_book;

NAME         PHONE_NO CITY
---------- ---------- ----------
Subodh     9897999999 hdr
Nish        807773435 Del
Raj         349235745 Hyd

SQL> delete from Phone_Book;

3 rows deleted.

SQL> select * from Phone_Book;

no rows selected
```